Breast Cancer Classification Using Neural Networks

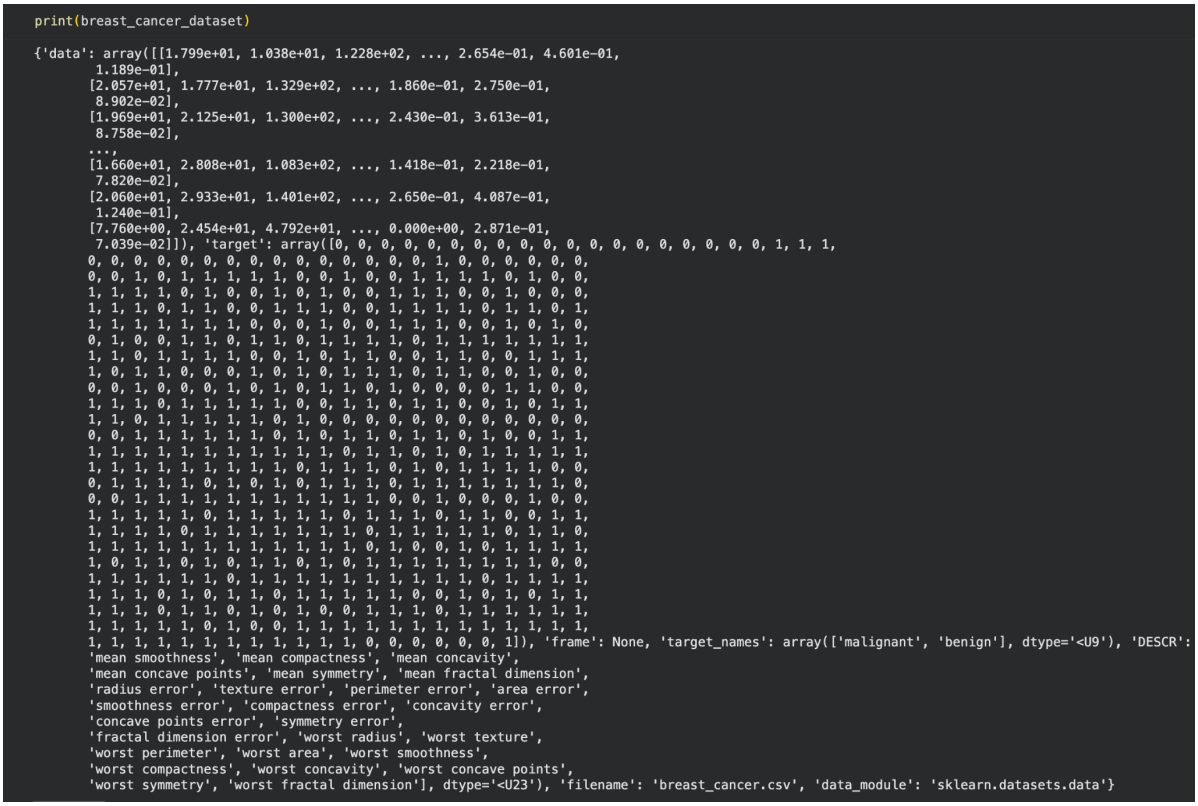
Project Execution Screenshots

AI & Machine Learning Internship – Naviotech Solutions

Overview

This document presents the key execution screenshots captured during the development of the Breast Cancer Classification project. The screenshots illustrate dataset exploration, preprocessing, model training, evaluation, and prediction stages performed using Python, TensorFlow, Keras, Pandas, and Scikit-learn.

Figure 1. Original Breast Cancer Wisconsin Dataset

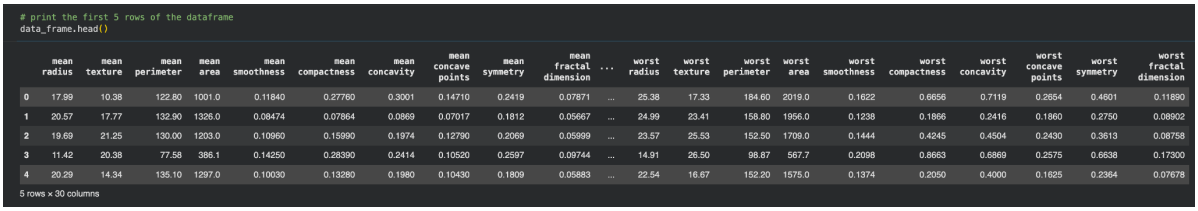


```
print(breast_cancer_dataset)

{'data': array([[1.799e+01, 1.038e+01, 1.228e+02, ..., 2.654e-01, 4.601e-01,
        1.189e-01],
       [2.057e+01, 1.777e+01, 1.329e+02, ..., 1.860e-01, 2.750e-01,
        8.902e-02],
       [1.969e+01, 2.125e+01, 1.300e+02, ..., 2.430e-01, 3.613e-01,
        8.758e-02],
       ...,
       [1.660e+01, 2.808e+01, 1.083e+02, ..., 1.418e-01, 2.218e-01,
        7.820e-02],
       [2.060e+01, 2.933e+01, 1.401e+02, ..., 2.650e-01, 4.087e-01,
        1.240e-01],
       [7.760e+00, 2.454e+01, 4.792e+01, ..., 0.000e+00, 2.871e-01,
        7.039e-02]]), 'target': array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 1, 0, 0,
       1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 0,
       1, 1, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
       1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1,
       0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 1, 1,
       1, 0, 1, 1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0,
       0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0,
       1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1,
       1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 0,
       0, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0,
       0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0,
       1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 1,
       1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1,
       1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0,
       1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1,
       1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1,
       1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1]), 'frame': None, 'target_names': array(['malignant', 'benign'], dtype='<U9'), 'DESCR':
       'mean smoothness', 'mean compactness', 'mean concavity',
       'mean concave points', 'mean symmetry', 'mean fractal dimension',
       'radius error', 'texture error', 'perimeter error', 'area error',
       'smoothness error', 'compactness error', 'concavity error',
       'concave points error', 'symmetry error',
       'fractal dimension error', 'worst radius', 'worst texture',
       'worst perimeter', 'worst area', 'worst smoothness',
       'worst compactness', 'worst concavity', 'worst concave points',
       'worst symmetry', 'worst fractal dimension'], dtype='<U23'), 'filename': 'breast_cancer.csv', 'data_module': 'sklearn.datasets.data'}
```

Figure 1. Original Breast Cancer Wisconsin Dataset

Figure 2. Dataset Preview (First Five Rows)



```
# print the first 5 rows of the dataframe
data_frame.head()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst radius	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	25.38	17.33	184.60	2019.0	0.1622	0.6656	0.7119	0.2654	0.4601	0.11890
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	24.99	23.41	158.80	1956.0	0.1238	0.1866	0.2416	0.1860	0.2750	0.08902
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	23.57	25.53	152.50	1709.0	0.1444	0.4245	0.4504	0.2430	0.3613	0.08758
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	14.91	26.50	98.87	567.7	0.2098	0.8663	0.6869	0.2575	0.6638	0.17300
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	22.54	16.67	152.20	1575.0	0.1374	0.2050	0.4000	0.1625	0.2364	0.07678

5 rows × 30 columns

Figure 2. Dataset Preview (First Five Rows)

Figure 3. Dataset Preview (Last Five Rows)

```
# print last 5 rows of the dataframe
data_frame.tail()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension	label
564	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	0.1726	0.05623	...	26.40	166.10	2027.0	0.14100	0.21130	0.4107	0.2216	0.2060	0.07115	0
565	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	0.1752	0.05533	...	38.25	155.00	1731.0	0.11660	0.19220	0.3215	0.1628	0.2572	0.06637	0
566	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	0.1590	0.05648	...	34.12	126.70	1124.0	0.11390	0.30940	0.3403	0.1418	0.2218	0.07820	0
567	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	0.2397	0.07016	...	39.42	184.60	1821.0	0.16500	0.86810	0.9387	0.2650	0.4087	0.12400	0
568	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	0.1587	0.05884	...	30.37	59.16	268.6	0.08996	0.06444	0.0000	0.0000	0.2871	0.07039	1

5 rows × 31 columns

Figure 3. Dataset Preview (Last Five Rows)

Figure 4. Dataset Shape

```
# number of rows and columns in the dataset
data_frame.shape
```

```
(569, 31)
```

Figure 4. Dataset Shape

Figure 5. Dataset Information

```
# getting some information about the data
data_frame.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 31 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   mean radius             569 non-null    float64
 1   mean texture            569 non-null    float64
 2   mean perimeter          569 non-null    float64
 3   mean area               569 non-null    float64
 4   mean smoothness         569 non-null    float64
 5   mean compactness        569 non-null    float64
 6   mean concavity          569 non-null    float64
 7   mean concave points     569 non-null    float64
 8   mean symmetry           569 non-null    float64
 9   mean fractal dimension  569 non-null    float64
 10  radius error            569 non-null    float64
 11  texture error           569 non-null    float64
 12  perimeter error         569 non-null    float64
 13  area error              569 non-null    float64
 14  smoothness error        569 non-null    float64
 15  compactness error       569 non-null    float64
 16  concavity error         569 non-null    float64
 17  concave points error    569 non-null    float64
 18  symmetry error          569 non-null    float64
 19  fractal dimension error 569 non-null    float64
 20  worst radius            569 non-null    float64
 21  worst texture           569 non-null    float64
 22  worst perimeter         569 non-null    float64
 23  worst area              569 non-null    float64
 24  worst smoothness        569 non-null    float64
 25  worst compactness       569 non-null    float64
 26  worst concavity         569 non-null    float64
 27  worst concave points    569 non-null    float64
 28  worst symmetry          569 non-null    float64
 29  worst fractal dimension 569 non-null    float64
 30  label                   569 non-null    int64
dtypes: float64(30), int64(1)
memory usage: 137.9 KB
```

Figure 5. Dataset Information

Figure 6. Missing Values Check

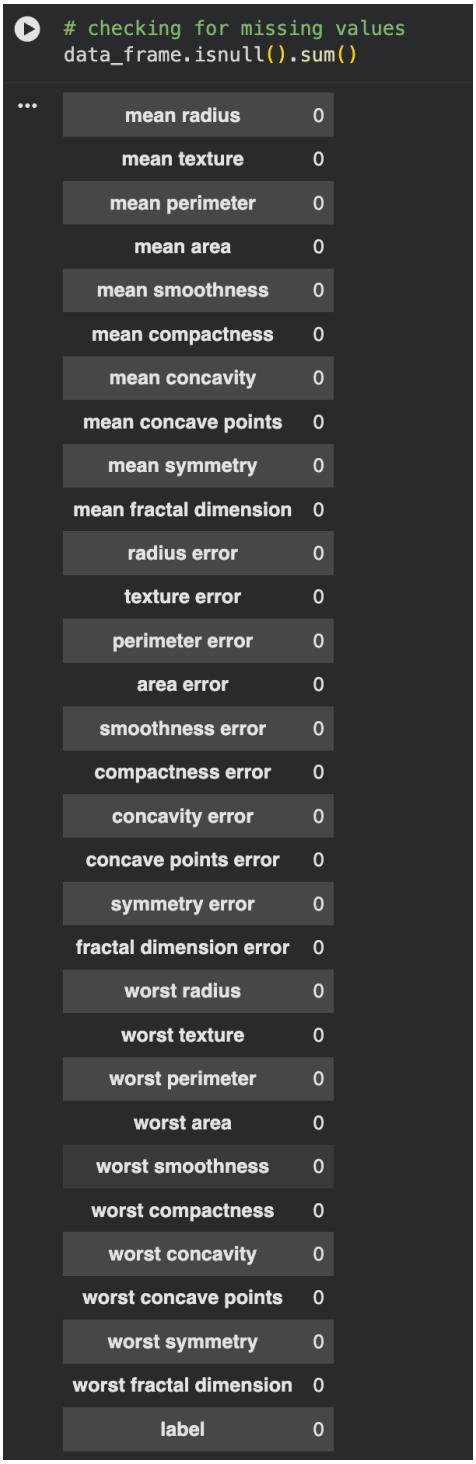


Figure 6. Missing Values Check

Figure 7. Statistical Summary

```
# statistical measures about the data
data_frame.describe()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension	label
count	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	...	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000	569.000000
mean	14.127292	19.289649	91.969033	654.889104	0.096360	0.104341	0.088799	0.048919	0.181162	0.062798	...	25.677223	107.261213	880.583128	0.132369	0.254265	0.272188	0.114606	0.290076	0.083946	0.627417
std	3.524049	4.301036	24.298981	351.914129	0.014064	0.052813	0.079720	0.038803	0.027414	0.007060	...	6.146258	33.602542	569.356993	0.022832	0.157336	0.208624	0.065732	0.061867	0.018061	0.483918
min	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.106000	0.049960	...	12.020000	50.410000	185.200000	0.071170	0.027290	0.000000	0.000000	0.156500	0.055040	0.000000
25%	11.700000	16.170000	75.170000	420.300000	0.086370	0.064920	0.029560	0.020310	0.161900	0.057700	...	21.080000	84.110000	515.300000	0.116600	0.147200	0.114500	0.064930	0.250400	0.071460	0.000000
50%	13.370000	18.840000	86.240000	551.100000	0.095870	0.092630	0.061540	0.033500	0.179200	0.061540	...	25.410000	97.660000	686.500000	0.131300	0.211900	0.226700	0.099930	0.282200	0.080040	1.000000
75%	15.780000	21.800000	104.100000	782.700000	0.105300	0.130400	0.130700	0.074000	0.195700	0.066120	...	29.720000	125.400000	1084.000000	0.146000	0.339100	0.382900	0.161400	0.317900	0.092080	1.000000
max	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.304000	0.097440	...	49.540000	251.200000	4254.000000	0.222600	1.058000	1.252000	0.291000	0.663800	0.207500	1.000000

8 rows × 31 columns

Figure 7. Statistical Summary

Figure 8. Target Variable Distribution

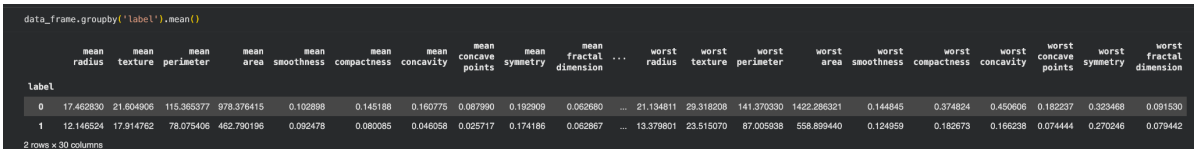
```
# checking the distribution of Target Varibale
data_frame['label'].value_counts()
```

label	count
1	357
0	212

dtype: int64

Figure 8. Target Variable Distribution

Figure 9. Grouped Feature Statistics



```
data_frame.groupby('label').mean()
```

label	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst radius	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension
0	17.462830	21.604906	115.365377	978.376415	0.102898	0.145188	0.160775	0.087990	0.192909	0.062680	...	21.134811	29.318208	141.370330	1422.286321	0.144845	0.374824	0.450606	0.182237	0.323468	0.091530
1	12.146524	17.914762	78.075406	462.790196	0.092478	0.080085	0.046058	0.025717	0.174186	0.062867	...	13.379801	23.515070	87.005938	558.899440	0.124959	0.182673	0.166238	0.074444	0.270246	0.079442

2 rows × 30 columns

Figure 9. Grouped Feature Statistics

Figure 10. Feature Matrix

```
X = data_frame.drop(columns='label', axis=1)
Y = data_frame['label']
```

```
print(X)
```

```
     mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0          17.99         10.38          122.80     1001.0          0.11840
1          20.57         17.77          132.90     1326.0          0.08474
2          19.69         21.25          130.00     1203.0          0.10960
3          11.42         20.38           77.58      386.1          0.14250
4          20.29         14.34          135.10     1297.0          0.10030
..           ...           ...             ...        ...              ...
564        21.56         22.39          142.00     1479.0          0.11100
565        20.13         28.25          131.20     1261.0          0.09780
566        16.60         28.08          108.30      858.1          0.08455
567        20.60         29.33          140.10     1265.0          0.11780
568         7.76         24.54           47.92      181.0          0.05263

     mean compactness  mean concavity  mean concave points  mean symmetry  \
0             0.27760         0.30010              0.14710         0.2419
1             0.07864         0.08690              0.07017         0.1812
2             0.15990         0.19740              0.12790         0.2069
3             0.28390         0.24140              0.10520         0.2597
4             0.13280         0.19800              0.10430         0.1809
..                ...             ...                  ...            ...
564           0.11590         0.24390              0.13890         0.1726
565           0.10340         0.14400              0.09791         0.1752
566           0.10230         0.09251              0.05302         0.1590
567           0.27700         0.35140              0.15200         0.2397
568           0.04362         0.00000              0.00000         0.1587

     mean fractal dimension  ...  worst radius  worst texture  \
0                   0.07871  ...        25.380          17.33
1                   0.05667  ...        24.990          23.41
2                   0.05999  ...        23.570          25.53
3                   0.09744  ...        14.910          26.50
4                   0.05883  ...        22.540          16.67
..                      ...  ...           ...            ...
564                 0.05623  ...        25.450          26.40
565                 0.05533  ...        23.690          38.25
566                 0.05648  ...        18.980          34.12
567                 0.07016  ...        25.740          39.42
568                 0.05884  ...         9.456          30.37

     worst perimeter  worst area  worst smoothness  worst compactness  \
0             184.60      2019.0           0.16220            0.66560
1             158.80      1956.0           0.12380            0.18660
2             152.50      1709.0           0.14440            0.42450
3              98.87       567.7           0.20980            0.86630
4             152.20      1575.0           0.13740            0.20500
..               ...         ...               ...                ...
564           166.10      2027.0           0.14100            0.21130
565           155.00      1731.0           0.11660            0.19220
```

Figure 10. Feature Matrix

Figure 11. Train-Test Split

```
print(X.shape, X_train.shape, X_test.shape)

(569, 30) (455, 30) (114, 30)
```

Figure 11. Train-Test Split

Figure 12. Neural Network Training

```
# training the Meural Network

history = model.fit(X_train_std, Y_train, validation_split=0.1, epochs=10)

Epoch 1/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 5s 55ms/step - accuracy: 0.6724 - loss: 0.6282 - val_accuracy: 0.8261 - val_loss: 0.4309
Epoch 2/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - accuracy: 0.8386 - loss: 0.4347 - val_accuracy: 0.9130 - val_loss: 0.2917
Epoch 3/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 1s 29ms/step - accuracy: 0.8875 - loss: 0.3297 - val_accuracy: 0.9348 - val_loss: 0.2207
Epoch 4/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - accuracy: 0.9120 - loss: 0.2662 - val_accuracy: 0.9565 - val_loss: 0.1809
Epoch 5/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9218 - loss: 0.2240 - val_accuracy: 0.9565 - val_loss: 0.1562
Epoch 6/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - accuracy: 0.9315 - loss: 0.1942 - val_accuracy: 0.9565 - val_loss: 0.1391
Epoch 7/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9438 - loss: 0.1719 - val_accuracy: 0.9565 - val_loss: 0.1263
Epoch 8/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9535 - loss: 0.1547 - val_accuracy: 0.9565 - val_loss: 0.1165
Epoch 9/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - accuracy: 0.9584 - loss: 0.1408 - val_accuracy: 0.9565 - val_loss: 0.1088
Epoch 10/10
13/13 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9609 - loss: 0.1293 - val_accuracy: 0.9565 - val_loss: 0.1026
```

Figure 12. Neural Network Training

Figure 13. Training Accuracy Curve

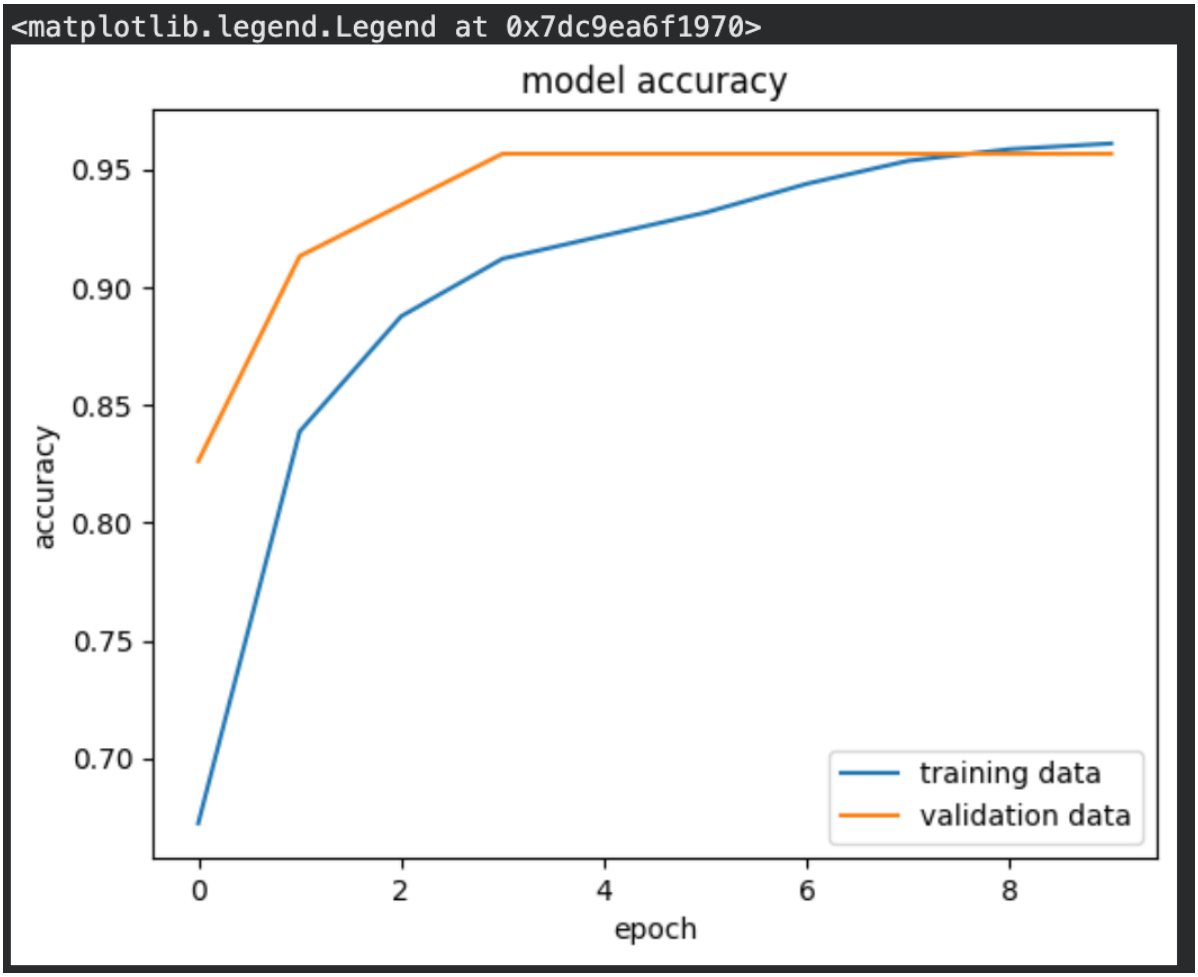


Figure 13. Training Accuracy Curve

Figure 14. Training Loss Curve

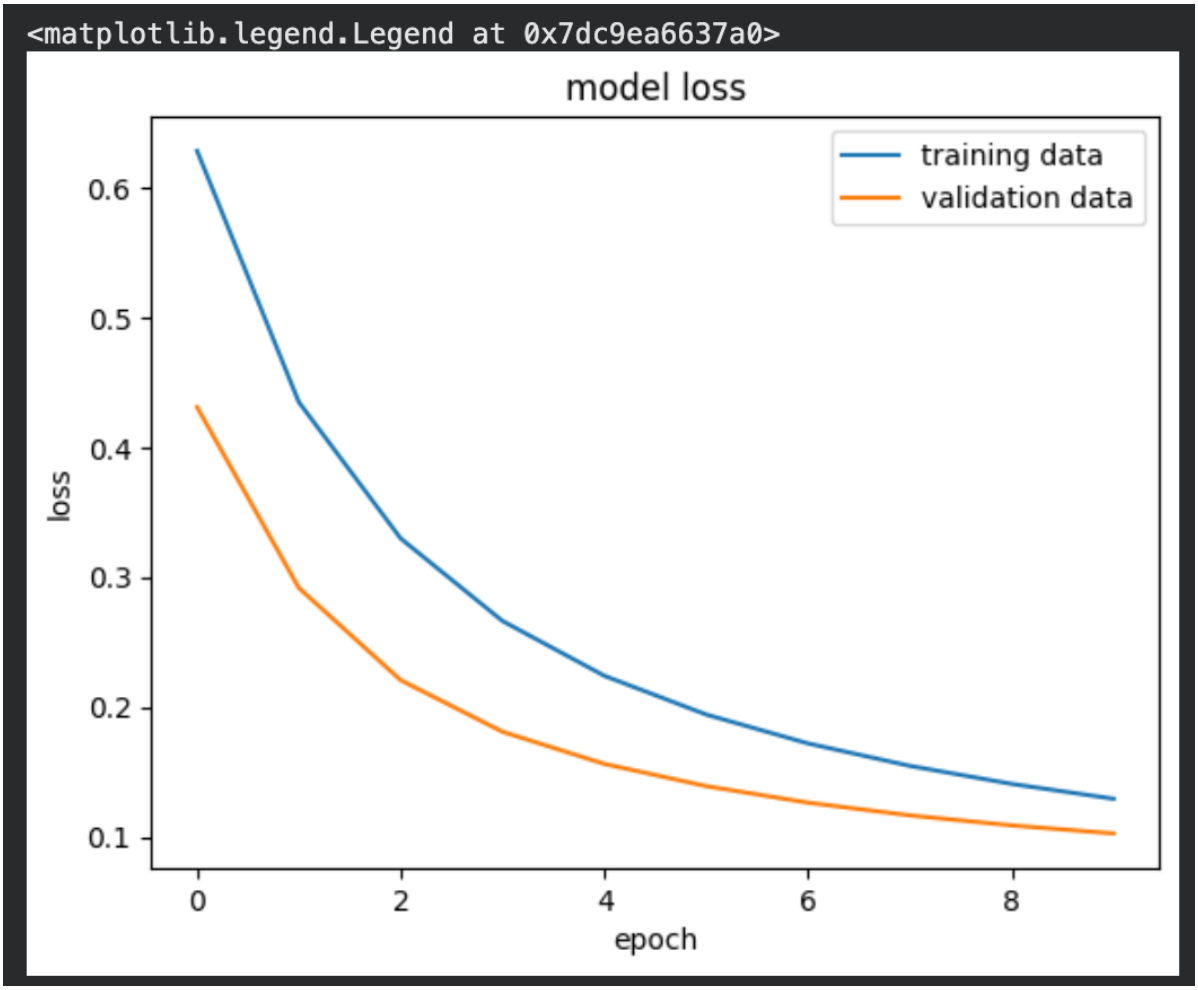


Figure 14. Training Loss Curve

Figure 15. Test Accuracy Evaluation

```
loss, accuracy = model.evaluate(X_test_std, Y_test)
print(accuracy)

4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - accuracy: 0.9386 - loss: 0.1440
0.9385964870452881
```

Figure 15. Test Accuracy Evaluation

Figure 16. Standardized Test Sample

```
print(X_test_std.shape)
print(X_test_std[0])
```

```
(114, 30)
[-0.04462793 -1.41612656 -0.05903514 -0.16234067  2.0202457  -0.11323672
  0.18500609  0.47102419  0.63336386  0.26335737  0.53209124  2.62763999
  0.62351167  0.11405261  1.01246781  0.41126289  0.63848593  2.88971815
 -0.41675911  0.74270853 -0.32983699 -1.67435595 -0.36854552 -0.38767294
  0.32655007 -0.74858917 -0.54689089 -0.18278004 -1.23064515 -0.6268286 ]
```

Figure 16. Standardized Test Sample

Figure 17. Prediction Generation

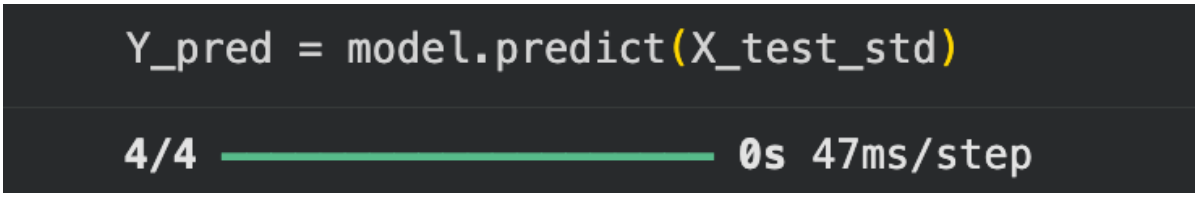


Figure 17. Prediction Generation

Figure 18. Prediction Shape

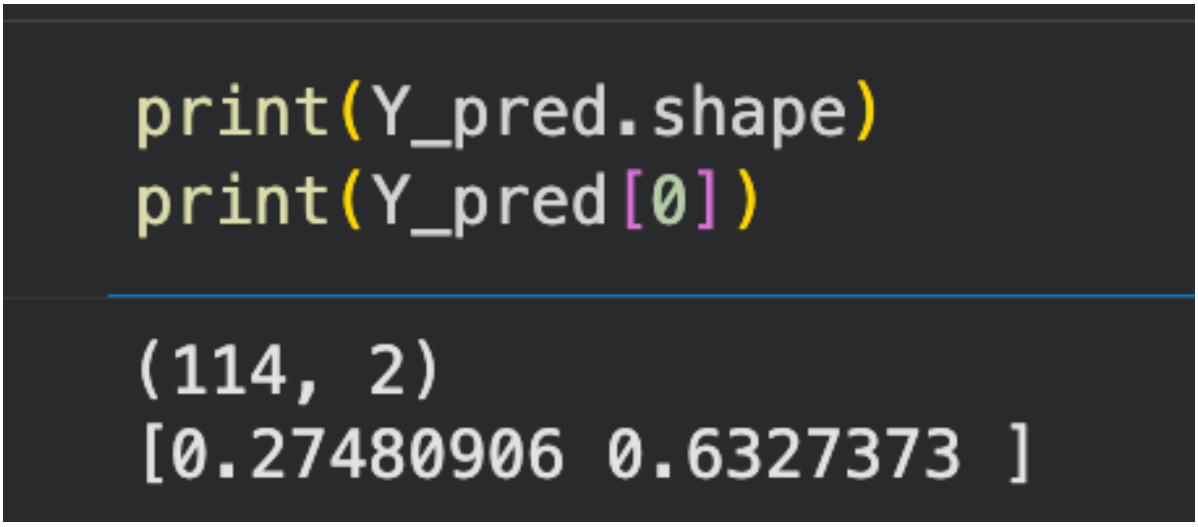


Figure 18. Prediction Shape

Figure 19. Prediction Probabilities

```
print(X_test_std)
```

```
[[-0.04462793 -1.41612656 -0.05903514 ... -0.18278004 -1.23064515
  -0.6268286 ]
 [ 0.24583601 -0.06219797  0.21802678 ...  0.54129749  0.11047691
   0.0483572 ]
 [-1.26115925 -0.29051645 -1.26499659 ... -1.35138617  0.269338
  -0.28231213]
 ...
 [ 0.72709489  0.45836817  0.75277276 ...  1.46701686  1.19909344
   0.65319961]
 [ 0.25437907  1.33054477  0.15659489 ... -1.29043534 -2.22561725
  -1.59557344]
 [ 0.84100232 -0.06676434  0.8929529  ...  2.15137705  0.35629355
   0.37459546]]
```

Figure 19. Prediction Probabilities

Figure 20. Argmax Demonstration

```
print(Y_pred)
```

```
[[2.74809062e-01 6.32737279e-01]
 [4.96673405e-01 4.83114421e-01]
 [1.06417097e-01 9.69048977e-01]
 [9.91918921e-01 6.94014737e-03]
 [4.88442034e-01 5.64689219e-01]
 [9.73612368e-01 1.52186500e-02]
 [2.68009990e-01 7.86602557e-01]
 [4.07565050e-02 9.37456548e-01]
 [1.95528999e-01 8.79152656e-01]
 [1.08843468e-01 8.33500564e-01]
 [5.03316581e-01 2.48250887e-01]
 [2.89251626e-01 7.66301095e-01]
 [1.23903498e-01 9.28007662e-01]
 [2.37132937e-01 7.69788802e-01]
 [1.60321370e-01 9.21853065e-01]
 [4.90640730e-01 2.40105838e-01]
 [1.33203432e-01 9.37348604e-01]
 [1.82419315e-01 8.97498608e-01]
 [2.86252290e-01 9.02042329e-01]
 [9.66008842e-01 5.40657304e-02]
 [8.90954614e-01 3.39664847e-01]
 [7.10288212e-02 9.45805311e-01]
 [1.67504773e-01 9.05270278e-01]
 [4.58845794e-02 9.03634071e-01]
 [1.49644181e-01 7.57499218e-01]
 [8.22138071e-01 8.86185244e-02]
 [1.21761613e-01 7.71392405e-01]
 [1.69083953e-01 5.95029891e-01]
 [7.44201422e-01 1.12871371e-01]
 [7.41491497e-01 7.06996992e-02]
 [1.35615170e-01 6.75073802e-01]
 [1.76433980e-01 8.45172048e-01]
 [1.84185967e-01 8.72628570e-01]
 [9.87049937e-01 4.08998516e-04]
 [9.64470983e-01 9.78886187e-02]
 [3.28604639e-01 8.52663398e-01]
 [3.34387869e-02 9.70259488e-01]
 [3.33620995e-01 7.21441388e-01]
 [4.56540324e-02 9.63423312e-01]
 [1.63606271e-01 8.91346514e-01]
 [9.96114850e-01 1.00097274e-02]
 [6.10745609e-01 1.91879272e-01]
 [7.98925757e-03 9.85343456e-01]
 [1.79544657e-01 9.31271255e-01]
 [8.02839518e-01 2.74308443e-01]
 [2.03447953e-01 8.75126362e-01]
 [1.08570494e-01 9.56016898e-01]
 [8.60047340e-02 9.63898063e-01]
 [8.90604436e-01 5.09861950e-03]
 [7.80323863e-01 1.15475856e-01]
 [1.09163642e-01 8.01459014e-01]
 [6.00945771e-01 2.88915485e-01]
 [2.24471688e-01 4.18986529e-01]
 [9.16811973e-02 9.22453105e-01]
 [6.92879781e-02 9.50322628e-01]
 [4.06329036e-01 4.25174803e-01]
 [6.68855831e-02 8.84578347e-01]
 [5.63205034e-02 9.75570083e-01]
```

Figure 20. Argmax Demonstration

Figure 21. Final Tumor Prediction

```
#  argmax function

my_list = [0.25, 0.56]

index_of_max_value = np.argmax(my_list)
print(my_list)
print(index_of_max_value)
```

```
[0.25, 0.56]
1
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
[[0.05689582 0.9499567 ]]
[np.int64(1)]
The tumor is Benign
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
  warnings.warn(
```

Figure 21. Final Tumor Prediction